

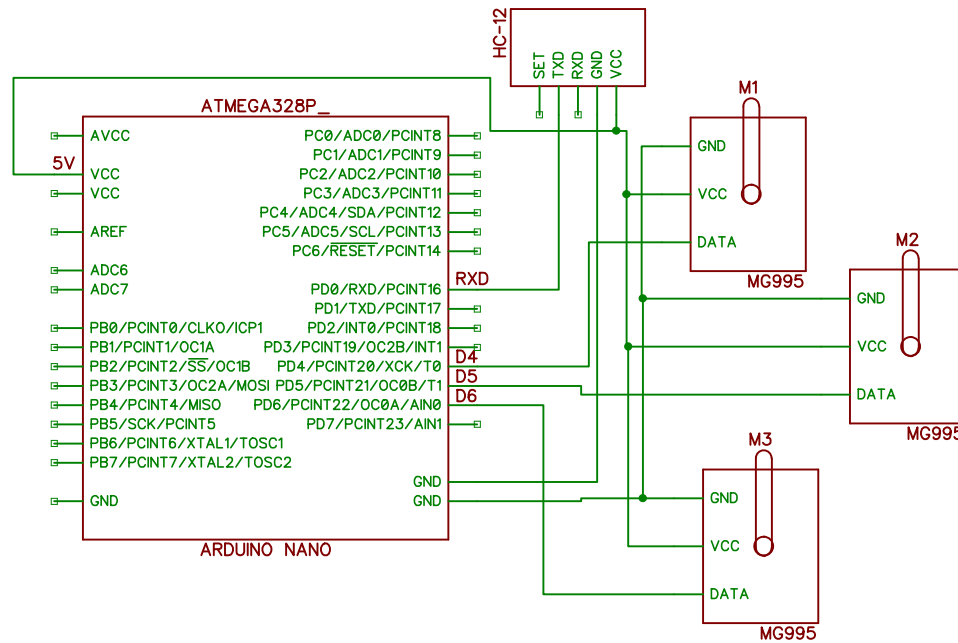
The diagram illustrates the internal structure and external connections of an ATmega328P microcontroller. The central component is the ATmega328P, with its pins labeled as follows:

- AVCC**: Analog supply voltage
- VCC**: Digital supply voltage (5V)
- AREF**: Analog reference voltage
- ADC6**, **ADC7**: Analog-to-digital converter pins
- PB0/PCINT0/CLKO/ICP1**, **PB1/PCINT1/OC1A**, **PB2/PCINT2/SS/OC1B**, **PB3/PCINT3/OC2A/MOSI**, **PB4/PCINT4/MISO**, **PB5/SCK/PCINT5**, **PB6/PCINT6/XTAL1/TOSC1**, **PB7/PCINT7/XTAL2/TOSC2**: Port B pins with various functions
- PC0/ADC0/PCINT8**, **PC1/ADC1/PCINT9**, **PC2/ADC2/PCINT10**, **PC3/ADC3/PCINT11**, **PC4/ADC4/SDA/PCINT12**, **PC5/ADC5/SCL/PCINT13**, **PC6/RESET/PCINT14**: Port C pins with various functions
- PD0/RXD/PCINT16**, **PD1/TXD/PCINT17**, **PD2/INT0/PCINT18**, **PD3/PCINT19/OC2B/INT1**, **PD4/PCINT20/XCK/T0**, **PD5/PCINT21/OC0B/T1**, **PD6/PCINT22/OC0A/AIN0**, **PD7/PCINT23/AIN1**: Port D pins with various functions
- GND**: Ground pins

The ATmega328P is connected to three Arduino boards:

- ARDUINO NANO (Left)**: Connected via P1 and P2 headers. P1 is connected to the VCC pin, and P2 is connected to the GND pin.
- ARDUINO UNO (Bottom)**: Connected via its pins. The VCC pin is connected to the 5V pin, and the GND pin is connected to the GND pin.
- ARDUINO NANO (Right)**: Connected via its pins. The VCC pin is connected to the VCC pin, and the GND pin is connected to the GND pin.

RECEIVER 1



RECEIVER 2

